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South2 Corridor; From Algeria to Germany; The Promising Future of Hydrogen Exports

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Abstract

This paper will showcase the importance of a great H₂ project as Europe is facing many challenges to find sustainable solutions to its energy needs; one of the key elements is to invest in green H₂. SouthH₂ Corridor is a hydrogen pipeline which is under feasibility study. It is supposed to transport green molecules from North Africa, starting from Algeria, passing through Tunisia, Italy, Austria, to Germany. The pipeline is 3300 km long and is able to supply 4 MT of green H₂ to Europe. In this paper, we would like to shed light on the huge contribution of this project to reduce a tremendous amount of carbon emissions as it will replace conventional resources in energy generation, namely coal.

Methods, Procedures, Process: SouthH₂ Corridor is not a simple project in the green hydrogen industry, but the first intercontinental project to transport green H₂ from North Africa to Southern Europe. An MOU was signed in October 2024 between the national companies Sonatrach and Sonelgaz from Algeria, VNG from Germany, SNAM and Sea Corridor from Italy, and Verbund Green Hydrogen from Austria.

As Europe is planning to produce 10 MT of green H₂ and import the same volume by 2030, the SouthH₂ project brings added value to Europe as it represents 40% of the total imports.

Algeria has the largest surface area in Africa, 90% of which is desert, making it a vital source of renewables for Europeans, as the geographic position and the abundance of solar energy are key factors that will help many European countries meet their carbon neutrality plans by 2050. That is why the SouthH₂ Corridor is considered a strategic project to lower carbon emissions by replacing coal, which is still largely used in Europe.

Results, Observations, Conclusions: SouthH₂ Corridor will bring green molecules from Algeria to Germany as the final destination. Bringing 4 MT of green H₂ will replace many fossil fuels in power generation, assuming that 1 MWh generated from coal emits 953 kg of CO₂, noting that from 1 MT of green H₂ we can generate 33 TWh of energy content. Therefore, 4 MT will deliver 132 TWh through SouthH₂ Corridor; billions of tons of CO₂ will be eliminated.

A detailed presentation will showcase the importance of this project as a pioneer in the world to transport green H₂ from one continent to another.

Novel/Additive Information: Technology and investments are key elements to implement this project and are available to reach this target. The project will make a revolution in the green H2 industry, as it will overcome many technical challenges, contributing to reducing production and transportation costs.

Introduction

Amid the global strategy to meet climate targets by reducing the carbon footprint, the world still relies on fossil energies for more than 80%, and these account for 75% of global greenhouse gas emissions and 90% of carbon dioxide emissions, according to a report by the United Nations.

Most countries have set targets to decarbonize many sectors or even cap carbon neutrality; China announced its strategy to reach its objectives by 2060, India by 2070, and Europe by 2050.

Although the mentioned targets seem overly ambitious and the deadlines tight, many processes and regulations have been gradually integrated to meet these objectives.

Europeans are showing more commitment to limiting carbon emissions, as several regulations and mechanisms have been prepared (some are under preparation or approval). CBAM (Carbon Border Adjustment Mechanism) and ETS (Emissions Trading System) are the most vital ones to date, as strict and tight instructions have driven economic operators to adhere to them in order to limit emissions gradually.

Applying ETS has been set to allow Europeans to meet their targets of a 55% reduction in GHG emissions by 2030 compared to 1990, and net zero emissions by 2050. Other regulations include MRV (Monitoring, Reporting and Verification) and MSR (Market Stability Reserve); all of them, however, have one single focus: reducing emissions.

Although targets are promising, Europeans still rely on fossil energies for more than 68.7% of their energy mix, as local investments in renewables still do not cover the majority of energy generation, which is a big challenge to meet the rising demand for energy.

One of the key solutions to overcome these constraints is to import green energies from another continent, i.e., Africa. In this paper, we will shed light on one of the largest promising green hydrogen projects in the world: SouthH2 Corridor; a pipeline of 3300 km which will connect North Africa starting from Algeria, passing through Tunisia, Italy, Austria, and ending in Germany (Fig. 1).

Project Name	South2 Corridor
Length	3300 km
Contributing countries	05 (Algeria-Tunisia-Italy-Austria-Germany)
Listed PCI (Project of Common Interest)	08 th April 2024
Planned supply capacity per annum	04 MTPA
Operational Plan	2030

Figure 1—South2 Corridor information

In October 2024, a memorandum of understanding was signed by these countries to study the feasibility of this project, which aims to produce green hydrogen in Algeria and partially in Tunisia from solar and wind energies to be transported through pipelines in a gaseous state to European partners.

Many companies have shown great interest in contributing to this project for many reasons, notably the short distance from North Africa to Southern Europe (South of Italy), the competitiveness of the produced green H2, the abundance of green energies which encourages investors, and the know-how to build pipelines from North Africa to Southern Europe as 3 pipelines of natural gas are already in place: two of them supply Italy with natural gas (Transmed), while the third supplies Spain (MedGaz), all originating from Algeria.

Adding to these encouraging parameters, Europeans are seeking to diversify their energy needs and are heavily subsidizing green projects as a priority to meet their climate targets. SouthH2 Corridor will not only connect one continent to another but will contribute to making Europe greener.

Scope and Objectives of South2 Corridor

In 2022, the EU launched a plan called REPower, through which Europe aimed to reduce its dependency on fossil fuels by accelerating the process for a vital energy transition; however, reaching this vision requires huge investments inside the continent but is still not enough to meet its targets as the industry still heavily relies on fossil fuels, with the lack or instability of energy supplies from wind and solar in many countries and in order to balance supply and demand.

One great project has been born in the last few months planning to support Europeans in their energy transition: SouthH2 Corridor (Fig. 2 – SouthH2 Itinerary), a pipeline of 3300 km which is under feasibility studies (an MOU was signed by 5 countries) focusing on supplying about 4 MTPA of green hydrogen to 3 main European countries (Italy, Austria, and Germany). Its main objective is to gradually replace fossil energies, particularly coal, as many countries still burn it, especially in the eastern part of the continent.



Figure 2—South2 Itinerary

In Germany, 22% of its electricity generation is ensured by coal; fossil fuels contributed 43% to electricity generation in 2024 in the country. Phasing out coal is a target of many countries in Europe, and here green hydrogen can largely contribute to taking an important share as Germany, for example, is planning to launch its H2-ready gas power plants to be operational by 2030.

Green hydrogen will not only support local economies, but also replace the most polluting fossil fuel, which is coal. The priority now is how to implement this project knowing that some challenges could be faced at the beginning, given that it is the first project in the world to transport green hydrogen from one continent to another.

Main objectives of the project

- Lessen the dependency on fossil energies
- Replace coal, especially in power generation
- Meet Europe's climate targets by limiting emissions

- Increase the share of renewables in the energy mix
- Encourage Europeans to invest in green hydrogen
- Enable European firms to acquire new expertise in hydrogen transportation
- Expand green hydrogen in industry and other sectors
- Include hydrogen energy in new businesses by expanding its uses
- Master the green hydrogen value chain
- Spread hydrogen derivatives utilization in various domains

Achieving the project's objectives will certainly face many challenges:

- Technical challenges: use of the existing natural gas infrastructure to transport green hydrogen (repurposing) or building a new pipeline (which requires large investments).
- Pipe embrittlement: this is one of the major issues hydrogen transportation faces, as the smallest molecule of hydrogen can leak rapidly and create sudden cracks in the pipeline body.
- Using the existing pipeline infrastructure risks impacting the steady supply of natural gas, which is still the backbone of many economies inside the Eurozone; this option could face some resistance from natural gas importers as it may damage their interests.
- The cost of the project.
- Renewables production mix: the share of solar and wind energy produced; the absence of solar rays at night is another gap that renewable energies have to fill to maintain steady electricity deliveries to the electrolysis and water desalination stations.
- The concept of delivering 100% green hydrogen or having flexibility like a hybrid system in harsh conditions (may use natural gas, being the cleanest among fossils, to deliver electricity to electrolysis and fresh water production plants).

Environmental Impact of South2 Corridor

Implementing SouthH2 Corridor will have positive consequences on the environment. Replacing energy sourced from fossils, especially coal, will certainly save millions of tons of carbon as Europe still heavily depends on fossils for power generation (Fig. 3). Green hydrogen is supposed to play a major role in the energy transition because of its carbon neutrality in the production process, its rich energy content, and its ability to be used in many applications.

The previous facts will drive hydrogen to be one of the main elements leading a smooth transition towards the global targets; that is why it is positioned as an important source of energy in the future regardless of its inconveniences like inflammation, transportation challenges, and even the repurposing cost of the available infrastructure (doable in some cases). Green hydrogen maintains a major position among renewables.

Even though the production cost is still not competitive enough compared to many sources of energy, large-scale investments will not only contribute to reducing the price but also will largely entice shareholders and investors to contribute to carbon limitations and thus winning a great deal, i.e., climate.

The success of the project will be a great jump and a new move towards clean and sustainable energies; technologies will play a pivotal role in reducing emissions as this will favor the expansion of new projects connecting plants and facilities with new sources of energy that are free of emissions and will certainly pave the way for tailored and targeted strategies of policymakers by setting new regulations and laws in favor of green hydrogen spread throughout the supply chain.

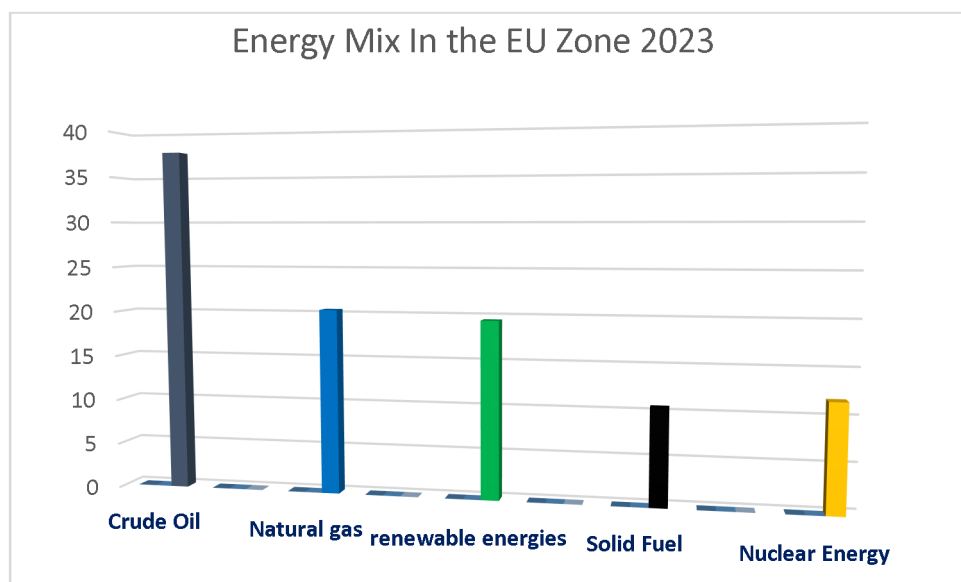


Figure 3—Energy Mix in Europe 2023

Carbon regulations will also be reformulated. Down below, we can have a look at carbon savings once connecting green hydrogen in replacement of some energy sources, such as coal.

Carbon saving: SouthH2 Corridor will bring green molecules from North Africa to Germany passing through Tunisia, Italy, and Germany. Green H2 molecules will contribute to eliminating millions of tons of carbon emitted into the air; those volumes cause pollution and even have negative consequences on European policy to reduce emissions. SouthH2 Corridor is expected to transport 4 Mt of green hydrogen annually to Europe. Assuming that 1 Mt of green H2 will generate about 33 TWh of energy, it will have approximately 132 TWh to be supplied per annum, which exceeds the electricity consumption in the Netherlands.

Europe is planning to produce 10 Mt of green H2 by 2030 and import the same volume; 4 Mt per annum from SouthH2 represents 40% of the imports, which is why this project is vital for the continent's energy transition plans and decarbonization of many sectors such as steel and aluminum, which are still emitting a considerable amount of CO2. Involving green H2 in the industry will be an added value to many regional economies to alleviate the emissions burden.

If we look at coal emissions, 1 MWh generated from coal emits approximately 953 kg of CO2, taking into account that the share of coal in electricity generation was estimated at 12% of the total electricity supplied in 2023 in the continent. Replacing 1 MWh of coal with renewable alternatives means that we save 953 kg of CO2; in 132 TWh we save more than 125 million tons of CO2, representing annually roughly 3.4% of Europe's emissions estimated at 3.6 billion tons (2023).

Green Hydrogen Production cost

The Levelized Cost of Hydrogen (LCOH) production in Europe ranges from 3.5 to 7 EUR/kg, which is still not competitive compared to North Africa. Solar energy production and even wind in Europe are not stable seasonally and monthly, which creates some perturbations in power outputs. The Dunkelflaute phenomenon largely impacted power generation in the winter of 2024-2025 and reduced wind energy supplies by 15%, which drove governments to rely on other alternatives like coal, natural gas, and heating oil. Solar energy is available but not at rates sufficient to generate large volumes to overcome local demand. Finding other trusted and vital alternatives on board became a priority to adhere to the European policy of net zero emissions by 2050.

One of the most competitive regions is Algeria, which has an abundance of solar energy all year round (90% of the surface area is desert) with an average sunshine of 8.2 hrs/day. This fact attracts investors and European partners to strengthen ties with local firms to ensure steady supplies of green electricity to generate green H₂ at competitive cost. These advantages include high production capacity (regular supplies).

Water potential to supply the project

Green hydrogen production requires huge volumes of fresh water; once talking about large-scale projects like SouthH₂ Corridor, it's crucially important to take into account how to provide a steady supply of fresh water to the production plants. But the advantages North African countries have are so considerable, and there is no issue about the sources, as the Mediterranean Sea is suitable to deliver the necessary mass of water responding to the demand. But this is not the only available option; recovering sewage water is considered one of the potential solutions to produce fresh water, as treatments are required to make fresh water. Desalination of the sea water will help eliminate the high content of salt. Afterwards, the required volumes will pass through electrolyzers to split water molecules into hydrogen and oxygen. A minimum of 9 liters of fresh water is needed to produce 1 kg of green hydrogen. For SouthH₂ Corridor, and by targeting 4 Mt of green hydrogen per annum, we need to treat at least 36 million cubic meters of fresh water (to be desalinated) to be amended to the project as a minimum, as other extra volumes will be used in the processing – leakage, etc. That's why recovering extra volumes from sewage or the valley water coming from mountains, especially in winter, will help reduce the treatment cost and make it more competitive in several regions where stations will be implemented. Combining all water sources together (sea water desalination, fresh water from valleys, and sewage water treatment) will diversify the water supplies and make it more flexible.

36 billion liters are required for this project; in the feasibility studies, these parameters (share of water sources) will be highlighted by the engineering companies involved in the project and will point out the share of the existing potentials. At the beginning, sea water will take the larger share but can be lowered gradually based on the cost of treatment and the investment dedicated to the two other options. The abundance of sea water is so advantageous to get a highly competitive LCOH, contributing to afford continuous supplies for different treatment stations in the North of Algeria. Noting that the national oil company Sonatrach is leading water desalination in the country, having long expertise in this field. More than 2.2 billion cubic meters are treated annually, providing fresh water mainly to households. The required water for SouthH₂ Corridor represents only 1.65% of the total desalinated water in the country. In Fig. 4, we see how this project is easily doable and achievable based on the great potentials already existing in the country, now representing only a small share of the annual supplies. The large-scale investment in desalination allowed mastering the processing of production, reducing cost, and making profit by improving the competitiveness of production, driving this project toward great success.

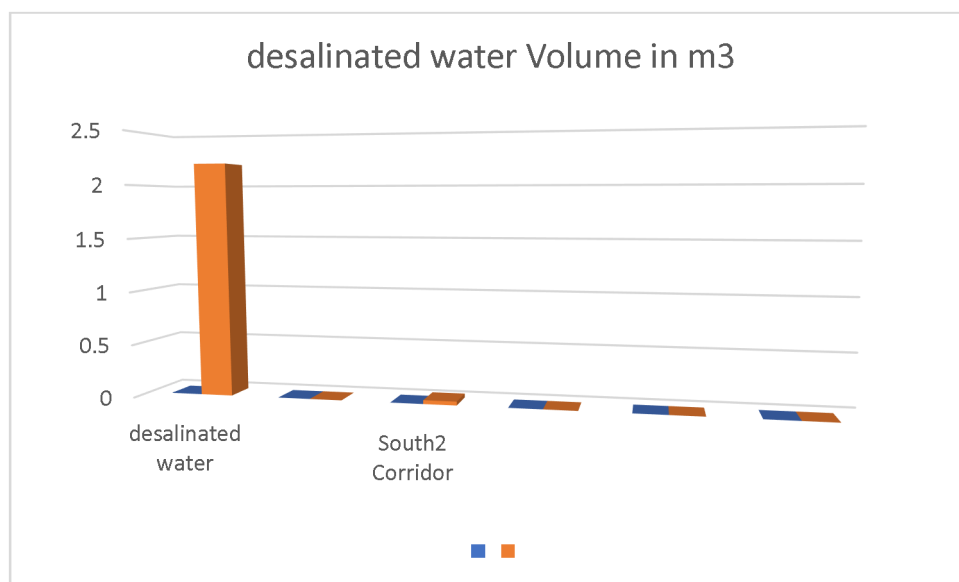


Figure 4—comparison between the existing desalination and south2 requirements

Technical analysis of the Project

SouthH2 Corridor will start from North Africa (north of Algeria) and will cross Tunisia; it will be directly connected offshore to Augusta, south of Italy. The studies are underway to shape the pipeline and define depths and compression stations. Even though green hydrogen is still a challenging scenario as its tiny molecules can easily escape, many studies have been conducted in many labs and scientific institutions to find the right material for transportation. There are several options to transport hydrogen, as blending is still a possibility but has some limitations because of the pressure thresholds. Increasing the pipeline pressure in the existing infrastructure of the natural gas pipeline already connecting north of Algeria to south Italy may affect the pipeline's resistance by damaging the mechanical properties of the infrastructure. This option could be used but has limited applications at the moment. The second scenario is to build a new pipeline with a specific design for SouthH2 Corridor. X70 steel, X60, and X50, for example, are some of the potential grades for hydrogen transportation, but the safety factor has to be taken into account. Increasing the pipeline thickness is also proportional to that factor. Managing the pressure in the H2 pipeline is key.

In October 2020, Corinth Pipeworks (CPW) delivered the first pipes for a 440 km high-pressure pipeline in Italy. This pipeline is designed to transport hydrogen at concentrations up to 100%. The pipes are made from L415ME (API 5L X60) carbon steel from ArcelorMittal and meet ASME B31.12 (see Fig. 5) Option B standard. The pipeline has a diameter of 26 inches. Let's highlight that the ASME B31.12 code provides comprehensive guidelines for the design, construction, testing, and operation of hydrogen piping and pipelines. It plays a crucial role in ensuring the safe handling and transport of hydrogen, a gas that presents unique challenges due to its properties, such as high diffusivity and the potential to cause material embrittlement.

Repurposing the existing pipelines requires very strict regulations and qualification specifications, as several samples have to be taken to laboratories for tests and analyses. The challenges look a bit difficult when dealing with offshore pipelines, as working conditions are different from those on the surface. Interventions are generally expensive and take time to execute.

ASME B 31.8

841 STEEL PIPE

841.1 Steel Piping Systems Design Requirements

841.1.1 Steel Pipe Design Formula

(a) The design pressure for steel gas piping systems or the nominal wall thickness for a given design pressure shall be determined by the following formula (for limitations, see [para. 841.1.3](#)):

(U.S. Customary Units)

$$P = \frac{2St}{D} FET$$

(SI Units)

$$P = \frac{2.000St}{D} FET$$

where

D = nominal outside diameter of pipe, in. (mm)

E = longitudinal joint factor obtained from [Table 841.1.7-1](#) [see also [para. 817.1.3\(d\)](#)]

F = design factor obtained from [Table 841.1.6-1](#). In setting the values of the design factor, F , due consideration has been given and allowance has been made for the various underthickness tolerances provided for in the pipe specifications listed and approved for usage in this Code.

P = design pressure, psig (kPa) (see also [para. 841.1.3](#))

S = specified minimum yield strength, psi (MPa), stipulated in the specifications under which the pipe was purchased from the manufacturer or determined in accordance with [paras. 817.1.3](#)

(h) and [841.1.4](#). The specified minimum yield strengths of some of the more commonly used piping steels whose specifications are incorporated by reference herein are tabulated for convenience in [Mandatory Appendix D](#).

T = temperature derating factor obtained from [Table 841.1.8-1](#)

t = nominal wall thickness, in. (mm)

ASME B 31.12

PL-3.7 STEEL PIPELINE

PL-3.7.1 Steel Piping Systems Design Requirements

(a) *Steel Pipe Design Formula.* The design pressure for steel gas piping systems or the nominal wall thickness for a given design pressure shall be determined by the following formula [for limitations, see (b) below]:

$$P = \frac{2St}{D} FETH_f$$

where

D = nominal outside diameter of pipe, mm (in.)

E = longitudinal joint factor obtained from [Table IX-3B of Mandatory Appendix IX](#)

F = design factor obtained from [Table PL-3.7.1-1](#) or [Table PL-3.7.1-2](#) as applicable, depending upon the fracture control option specified in (b) below used in the design. In setting the values of the design factor, F , due consideration has been given and allowance has been made for the various underthickness tolerances provided for in the pipe specifications listed and approved for usage in this Code.

H_f = material performance factor from [Mandatory Appendix IX, Table IX-5A](#). Material performance factors account for the adverse effects of hydrogen gas on the mechanical properties of carbon steels used in the construction of pipelines.

P = design pressure, kPa (psig) [see also (b) below]

S = SMYS, kPa (psi), stipulated in the specifications under which the pipe was purchased from the manufacturer or determined in accordance with (c) below. The SMYS of some of the more commonly used pipeline steels whose specifications are incorporated by reference herein are tabulated for convenience in [Mandatory Appendix IX, Table IX-1B](#).

T = temperature derating factor obtained from [Table PL-3.7.1-3](#)

t = nominal wall thickness, mm (in.)

Figure 5—ASME Code for H2 Transportation in pipelines

Conclusion

SouthH2 Corridor is a very promising project to deliver 20% of the European green hydrogen needs by 2030. A memorandum of understanding has been signed between companies from Algeria (Sonatrach and Sonelgaz), VNG from Germany, SNAM and Sea Corridor from Italy, and Verbund Green Hydrogen from Austria; this took place in Oran in October 2024. The feasibility studies are underway to complete the first phase of the project evaluation. The potentials for the production of both renewable electricity and green hydrogen are crucially important and attractive with a very competitive price. Pipelines for hydrogen transportation are available with high pressure ratings able to support high loads. The project is under technical and financial evaluation, as meetings between the firms mentioned above are frequent to follow up on the procedures. After achieving this phase, a proposal for the financial estimation will be submitted for review and approval. SouthH2 Corridor will not only transport green molecules from North Africa to Europe but will also bring new horizons to global green hydrogen production and transportation by overcoming several technical hurdles long lived in the hydrogen industry.

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